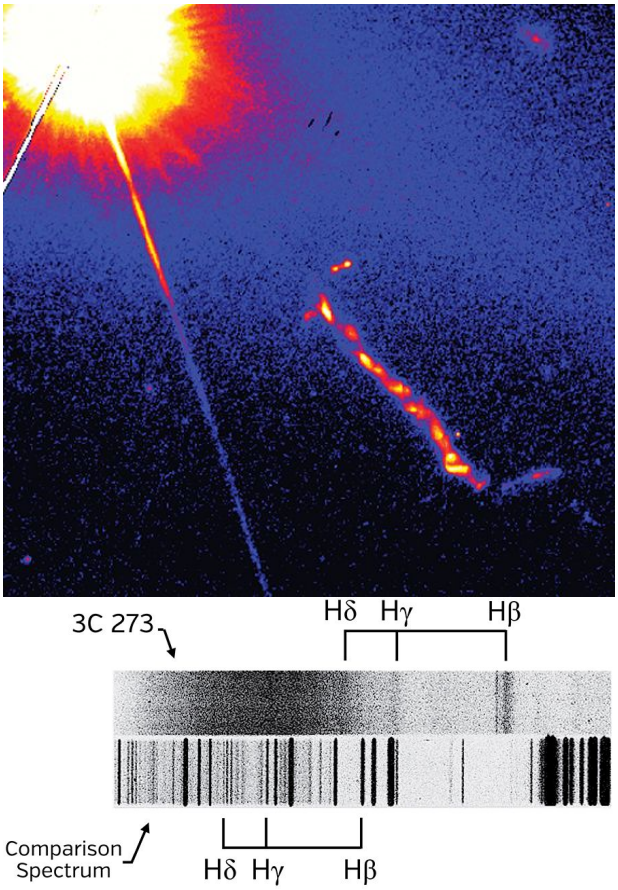


Fig. 9.1. *Top*: Photograph of 3C273 taken by NASA's Hubble Space Telescope. The star (*upper left*) looks big only because the photo is overexposed in order to see the faint jet (*lower right*). It is actually so small that its size cannot be measured. *Bottom*: Maarten Schmidt's spectral lines from 3C273 (*upper panel*) compared with spectral lines of hydrogen measured in an Earth laboratory. The quasar's three lines are the same as hydrogen's lines called Hβ, Hγ, and Hδ, but with wavelengths increased by 16 percent. (The images of the spectral lines are photographic negatives: black lines are really bright.)

As the universe expands, objects far from Earth move apart from us at very high speed, and objects nearer move away more slowly. 3C273's